

KYMA Framework Deep-Dive Dossier

A research-anchored plausibility map for a “zero-hardware” thought-to-text engine, with GOS/Honkular overlay for worldbuilding.

Executive summary

Your KYMA draft is most compelling (and easiest to make *technically convincing*) when treated as a **stack of three layers**:

Layer A: Real DSP/ML primitives (solid engineering).

Kalman/state-space tracking originates in classic optimal filtering. ¹

Welch PSD is a standard FFT-based spectral estimator. ²

FRFT is a real time-frequency generalization that is used in signal processing, including radar vibrometry contexts. ³

Vector quantization (VQ-VAE) is a standard discrete representation strategy. ⁴

BART is a standard seq2seq transformer pretraining approach that can decode structured latents into coherent language. ⁵

Layer B: Real but “frontier” sensing (partially feasible).

WiFi CSI sensing is well established for device-free perception (movement, gestures, respiration), but it is not currently a validated channel for reconstructing **semantic thought**. The field’s scope and limits are summarized in a heavily cited CSI survey. ⁶

Your strongest real-world cousin for “contactless speech intent” is **silent speech via RF**, where the physical signal is *not thought*, but **lip/throat micro-motion**. The most relevant recent example is **Lip-Siri**, which targets open-vocabulary silent speech using Wi-Fi backscatter plus a specialized tag to separate reflections. ⁷

Your “Neuro-Wideband / WUKONG” idea aligns with current work that uses deep learning to extrapolate “extended CSI” from narrowband CSI, aimed at improving multipath resolution for sensing. ⁸

Your multimodal fusion framing is also well grounded: a recent multi-modal Wi-Fi sensing overview explicitly argues that other modalities can act as teachers or complementary signals. ⁹

Thermal + rPPG fusion can detect latent mental states (stress vs other affective shifts) using passive imaging; this supports KYMA’s idea of **context gating** even if it cannot decode full text. PeerJ ¹⁰ published “Your blush gives you away” showing strong early-fusion improvements. ¹¹

Layer C: Mythic physics/biology overlay (best handled as a diegetic theory).

Photosynthetic exciton transport research provides a scientifically literate “algae bridge” via coherence signatures and **environment-assisted quantum transport (ENAQT)**-style motifs. ¹²

Microtubule quantum consciousness (Orch-OR) is an active controversy: Tegmark’s decoherence analysis argues microtubule-scale quantum coherence decoheres extremely fast in the brain, while Orch-OR reviews defend a quantum role and have generated rebuttals and counter-rebuttals. This debate is ideal “hard-sci-fi scaffolding” precisely because it is unresolved. ¹³

The **key plausibility move** is to present KYMA not as “RF reads thoughts,” but as:

KYMA reads weak, involuntary outputs of cognition (subvocal/laryngeal micro-motion, autonomic signatures, attention/effort proxies)
+ **discretizes** them (codex / VQ)
+ **decodes** them with a language model under heavy priors and self-consistency constraints.

That makes the story’s technical core feel real, while leaving room for your GOS constants and recursion mechanics as the in-universe explanation for why the system “works too well.”

KYMA as an engineering system

Technical primer for your code modules and their closest real analogs

Your posted implementation reads like a **simulation-first “runtime”** that already contains the right *shape* of a real system (features → state estimation → discrete tokens → text). The semantics of each module map cleanly onto established methods:

State extraction and stabilization

- Your 8D state estimator is a classic linear-Gaussian tracking pattern, conceptually aligned with the original linear filtering formulation. ¹
- Your Welch PSD hook is a standard way to estimate dominant frequency / spectral centroid / flatness from short windows; Welch’s method is foundational. ²
- Your “PLL / lock quality / jitter” metrics match standard phase-lock loop ideas; PLLs are a core way to estimate and track phase/frequency in noisy environments. ¹⁴
- Your use of phase synchrony/PLV as a coherence gate has a tight real-world match: phase-locking statistics/PLV were proposed as a way to quantify phase covariance between brain signals with workable time resolution. ¹⁵

Time-frequency separation via FRFT

- FRFT is a real generalization used to represent chirped or LFM-like structure compactly; it is widely surveyed in signal processing. ¹⁶
- FRFT is also used in radar vibration estimation contexts; this is a direct bridge to your “phononic / Doppler tomography” metaphors. ¹⁷

Discrete codex / codebook quantization

- Your codebook quantizer is the conceptual cousin of VQ-VAE: encode continuous features to discrete indices by nearest-neighbor in an embedding space, then decode. ⁴
- DeWave is the most relevant “anchor paper” for your draft’s *specific* claim (“discrete codex encoding” for EEG-to-text); it uses a quantized encoder and alignment to a pretrained language model, and reports BLEU-1 values that your draft references as a baseline. ¹⁸

Language decoding

- Your narrative calls for a BART-style decoder; BART is a denoising seq2seq transformer with an encoder-decoder structure suitable for mapping structured latents to text. ⁵

Plausibility assessment of this software layer

This layer is largely **feasible** as software engineering—what changes is *what the inputs can truly support*.

Feasible (today) - The end-to-end *architecture skeleton* (feature extraction → state estimator → discrete quantization → LM decoding) is consistent with mainstream DSP+ML. ¹⁹

- You can absolutely implement real-time PSD, FRFT transforms, Kalman tracking, and token quantization on commodity hardware.

Partially feasible - Real-time decoding to meaningful text depends on (a) stable sensors and (b) a constrained task. DeWave shows EEG-to-text works *in a particular experimental setting* (ZuCo reading tasks) but this is not “free thought.” ²⁰

Speculative - “Open-vocabulary thought-to-text” without invasive neural signals remains speculative, because scalp EEG is low SNR for rich linguistic content, and ambient RF sensing measures *motion/propagation changes*, not neural firing.

Key empirical gaps - Demonstrating that your token stream correlates with *linguistic structure* more strongly than with confounds (attention, muscle tension, breathing, posture).

- Cross-subject generalization without per-user calibration (still hard even for EEG-to-text). ²⁰

Fictional mechanism that minimally violates physics - Treat the codex not as “direct thought symbols,” but as **latent cognitive mode codes** (attention/memory/inner-speech) that *strongly constrain* what text is plausible next. The LM does the heavy lift, and “ $\Psi(t)=1$ ” becomes a narrative constraint solver (self-consistency) rather than a claim about new physics.

Passive sensing layer

WiFi CSI, backscatter silent speech, and the SAR vibrometry bridge

Technical primer

What CSI is (in the real world)

WiFi Channel State Information (CSI) is a per-subcarrier complex channel estimate (amplitude/phase) that changes with multipath geometry and motion. CSI sensing uses these changes to infer human activity, positioning, vital signs, etc., with substantial literature on signal processing pipelines and limitations. ⁶

Why “thought” is not the measured quantity

Crucially: WiFi sensing measures **propagation changes** (movement, scattering, reflections). Any neuro-cognitive influence would need to manifest via something physical: micro-movement, muscle activation, breathing patterns, autonomic vascular changes, etc.

Closest match to “silent speech” in RF

Lip-Siri is directly relevant because it targets *silent speech* not by reading neural activity, but by extracting **lip-motion traces** from WiFi backscatter. It explicitly uses a frequency-shifted backscatter tag to isolate the reflections of interest from other motion. ⁷

Why WUKONG / Neuro-Wideband matters

Neuro-Wideband WiFi sensing argues that limited bandwidth caps multipath resolution, and proposes a learning approach to extrapolate an “extended CSI” representation (eCSI), improving sensing performance without truly measuring wider bandwidth. This is an excellent “KYMA-adjacent” technical idea because it gives you a plausible mechanism for better micro-motion resolution without magical antennas. ⁸

SAR vibrometry and your “phononic extraction” metaphor

Your “Biondi Doppler Tomography” inspiration is real in the sense that micro-motion/vibration signatures can be extracted in radar contexts; Biondi’s work discusses vibrations embedded in Doppler diversity and tomography settings. ²¹

Even more on-the-nose: there is literature explicitly describing FRFT-based vibration estimation in SAR, linking your FRFT module to vibrometry practice. ¹⁷

Assessment of KYMA’s CSI-based “passive neural transduction” claim

Claim: “WiFi CSI captures the ‘shadow’ of neural oscillations.”

- **Plausibility rating: speculative → implausible (as written).**

CSI can capture micro-motions (breathing, gestures, lip motion), but “neural oscillations directly” are not presently a validated physical target for commodity WiFi CSI. The plausible route is indirect: cognition → muscle activation / micro-motion → CSI.

Minimal-violation fictional mechanism

Make it explicit in-universe that **KYMA’s “neural” signal is actually a biomechanical proxy** (subvocal/laryngeal micro-motion, cranial micro-motion, vascular pulsation), and that “GOS” is the *interpretive math* that turns these into stable symbolic codes.

Open empirical gaps - Separating subvocal/lip/throat micro-motion from respiration, posture shifts, background motion, and multipath churn (Lip-Siri solves this partly with a tag). ²²

- Demonstrating robust generalization across rooms and antenna geometries (CSI surveys emphasize robustness challenges). ²³

Suggested experimental validations and failure modes

Validation path (realistic lab steps) 1. Pick a measurable target first: **respiration** or **lip motion** (not thought). CSI literature supports these as feasible. ²³

2. Add teacher modalities (camera, IMU, mic) for ground truth per multi-modal WiFi sensing practice. ²⁴

3. Only then test “inner speech” as a classification problem (e.g., “speaking silently” vs resting), not open vocabulary.

Common failure modes - Multipath nonstationarity (room changes break models). ²³

- “Signal leakage”: the model learns posture or breathing artifacts instead of the target phenomenon.

- Overclaiming semantic decoding when the measurable is a proxy.

Brain-to-text layer

DeWave, ZuCo, and what “open vocabulary” really means in EEG science

Technical primer

ZuCo as a canonical dataset family

ZuCo (Zurich Cognitive Language Processing Corpus) combines EEG and eye-tracking from subjects reading natural text; the dataset is explicitly positioned to study reading and language processing and has become a benchmark base for EEG-to-text approaches. ²⁵

DeWave as your main anchor

DeWave proposes discrete coding (“codex”) for EEG-to-text and reports BLEU and ROUGE results on ZuCo, including marker-free settings at lower performance. This is the most direct real correlate to your “discrete codex + BART” storyline. ¹⁸

Other EEG-to-text / open vocabulary pipelines

There is a broader family of open-vocabulary EEG-to-text approaches using pretrained language models, often evaluated on ZuCo v1/v2; these typically require careful representation learning and still live far from unconstrained “thought streaming.” ²⁶

Why invasive neuroprostheses are a different regime

High-performance “speech neuroprosthesis” results (attempted speech decoded to text) are typically achieved with implanted neural recordings, not passive RF. A 2023 Nature paper highlights advances and remaining challenges for unconstrained sentence decoding from neural activity (in a clinical context). ²⁷

KYMA claim evaluation: “DeWave + BART yields open-vocabulary thought-to-text”

Claim, decomposed - Part 1: “Discrete codex helps generalize and reduces marker dependence.”

Rating: partially feasible (supported directionally by DeWave). ²⁸

- Part 2: “Open-vocabulary decoding.”

Rating: partially feasible in constrained experimental tasks (e.g., reading paradigms) but not equivalent to “free thought.” ²⁰

- Part 3: “RF-extracted neural waves feed DeWave.”

Rating: speculative / implausible unless you explicitly redefine the RF signal as a proxy (lip/throat micro-motion, etc.) rather than EEG-like brain signals.

Comparison table: real methods vs KYMA claims

Capability	Representative real method	Typical latency envelope	Signal SNR reality	Hardware required	KYMA claim version	Plausibility rating
EEG-to-text on reading tasks	DeWave on ZuCo	Not primarily optimized as <300 ms real-time; research pipelines often offline	Low SNR; strong subject variability	EEG + dataset-specific preprocessing	"Raw wave → codex → BART"	Partially feasible 20
Open-vocabulary silent speech	Lip-Siri	Research prototype; depends on sensing frame rate	Depends heavily on isolating reflections	WiFi + backscatter tag	"WiFi reads inner speech micro-motion"	Partially feasible only with proxy + isolation 22
Multi-modal mental state detection	rPPG + thermal early fusion	Can be near real-time depending on compute	Autonomic proxies can be robust	Camera/thermal	"Thermal+acoustic as ground truth teacher"	Feasible as gating/context 29
Neural attempted speech decoding	Clinical neuroprosthesis	Approaching streaming; still complex	High SNR vs EEG	Implants	"Bridgelike performance without hardware"	Implausible without new physics/sensors 27

Multi-modal fusion layer

Thermal tomography, rPPG, and acoustic/PMUT sensing as "biological ground truth"

Technical primer

Thermal + rPPG as passive physiological readout

The "Your blush gives you away" work is a clean example of how autonomic measures (heart-rate-derived features and skin temperature dynamics) detect subtle mental state shifts, and how early fusion can outperform single-modality models. 30

Wi-Fi CSI thermal tomography (ESP32 boundary arrays)

A 2026 preprint proposes 2D indoor temperature field reconstruction using boundary ESP32 CSI links and a

tomography-style inverse problem; it explicitly reports an end-to-end pipeline runtime on the order of hundreds of milliseconds (310 ms) on a Raspberry Pi 4 in their setup. ³¹

PMUT sensing realism

PMUT technology is real as a sensing/actuation platform, with active research across imaging and sensing domains. ³²

A 2025 Nature Communications article describes a machine-learning-assisted wearable sensing system using PMUTs, reporting high sensitivity and wide bandwidth and classification performance “exceeding 96%” in their context—supporting your “PMUT + ResNet” motif, though not necessarily the exact 99.8% figure you quoted. ³³

KYMA claim evaluation: “Thermal + acoustic fusion gets 90%+ and sub-300 ms”

- **Fusion for robust classification of coarse cognitive states: feasible** as a research direction, with multiple demonstrations that fusion beats unimodal inference (thermal+rPPG; multi-modal WiFi sensing surveys). ³⁴
- **Room-scale CSI thermal tomography: plausible**, but note it produces **coarse spatial thermal fields**, not pinpoint “metabolic signature of subvocal intent.” Still, it’s a strong *worldbuilding plausibility anchor* because it uses commodity WiFi/ESP32 and real inverse problems. ³¹
- **“Zero-hardware” plus skin-attached PMUT sensors: contradiction by definition** (skin-attached is hardware). The clean narrative fix: call it **“zero-dedicated hardware”** (the environment already has sensors, or the PMUT patch is optional).

Open empirical gaps and failure modes

- Thermal and rPPG are sensitive to lighting, motion artifacts, and camera placement; misalignment can dominate. ²⁹
- CSI thermal tomography is room/geometry-dependent; inverse problems can be ill-conditioned without careful regularization. ³¹
- “90%+” claims depend on task definition; “binary stress classification” is not “open vocabulary thought-to-text.”

Optimization and “quantum-inspired” layer

QUBO/Ising mapping, quantum annealing, and what it can realistically do for KYMA

Technical primer

What QUBO is

Quadratic Unconstrained Binary Optimization (QUBO) encodes a wide range of combinatorial problems as minimizing a quadratic form over binary variables; it can be mapped to an Ising Hamiltonian form, enabling solution attempts via quantum annealing or other heuristics. ³⁵

What D-Wave Quantum ³⁶ hardware contributes

D-Wave’s systems and documentation emphasize that real quantum processors have specific connectivity graphs; Zephyr is a documented topology, and D-Wave provides topological and embedding materials that constrain practical QUBO problem sizes and structure. ³⁷

KYMA claim evaluation: “QUBO collapses the thought state space to text with sub-300 ms latency”

Feasible use-cases (today) - Feature selection / structured constraint satisfaction: Using QUBO as a constraint layer (select a consistent set of features/tokens) is plausible. ³⁵

- **Graph-structured smoothing:** A QUBO can encode “pick a token path that obeys constraints,” similar to Viterbi-like decoding but with different optimization machinery (often still better done classically unless problem structure fits hardware).

Partially feasible - Hybrid quantum-classical pipelines: In practice, D-Wave usage often involves hybrid solvers and careful embedding; speedups are problem-dependent and not guaranteed. ³⁸

Speculative - “Collapse the probability distribution faster than classical” as a general statement is speculative; quantum advantage is not automatic and depends heavily on instance families and overhead.

Minimal-violation fictional mechanism - Have KYMA’s QUBO layer serve as a **hard constraint satisfier** rather than an oracle: it enforces “ $\Psi(t)=1$ ” by selecting among multiple candidate token paths produced by a fast LM, pruning contradictions, and stabilizing the narrative stream.

Biology layer for the “Resonome” and algae bridge

Quantum transport in photosynthesis, microtubules, and gene-time realism

Quantum transport and your algae motif

The strongest scientifically grounded “algae quantum computer” bridge is not “FTL” or “soul exteriorization,” but this real theme:

- **Coherence signatures in photosynthetic energy transfer** were experimentally reported in landmark work (Engel 2007) and then explored at physiological temperature (Panitchayangkoon 2010). ³⁹
- **Non-equilibrium vibrational structure sustaining/recohering electronic coherence** is discussed in Chin 2013, which fits your “structured bath / tuned noise” motif. ⁴⁰
- **Environment-assisted quantum transport (ENAQT)** is a recognized theoretical phenomenon where noise can *improve* transport in open quantum systems; you can safely treat “7/4” as a story-specific *noise-coherence match point* that evolution discovered. ⁴¹

Orch-OR and microtubules as a contested “engine room”

Your draft uses Tegmark’s decoherence estimate as an obstacle and proposes “discrete decoherence shielding.” In the real literature:

- Tegmark’s analysis argues brain-relevant degrees of freedom decohere far too quickly for quantum computation to underlie cognition, including analysis of microtubule-related excitations. ⁴²
- Orch-OR reviews articulate the contrary view and situate the proposal in physics/biology/cosmology, but the theory is widely contested and has published critiques. ⁴³

- Microtubule lattice structure details (protofilament number, helix starts, lattice type discussions) have a deep structural biology literature; for example, classic work discusses helix starts and protofilament-number variation. ⁴⁴

Plausibility rating: speculative (and ideal for SF). The *debate* can be treated as real, while KYMA's "GOS resolution" becomes the fictional breakthrough.

Genomic Resonome: making it rigorous without "genes resonate at 140 Hz"

Your draft assigns Hz "OpCodes" to genes (FOXP2, CLOCK, ATP2C2, etc.). Real genomic biology supports:

- FOXP2 is strongly associated with speech and language phenotypes and is widely discussed as a key gene in speech/language disorders. ⁴⁵
- ATP2C2 has published associations with language impairment and related measures in genetic studies. ⁴⁶
- CLOCK/circadian gene networks operate on ~24 hour cycles, not 100 Hz; circadian clocks are transcription-translation feedback oscillators. ⁴⁷
- Transcription and protein expression dynamics live on seconds-minutes-hours timescales (bursting and reporter studies), not 0-200 Hz oscillations. ⁴⁸

Plausibility rating of "gene Hz resonances": implausible (as literal biology).

Minimal-violation fictional mechanism: reinterpret "gene resonances" as **priors and constraints** learned from genomics: - "FOXP2 mode" = a latent variable controlling speech-motor planning priors (language network gain). - "ATP2C2 mode" = a latent variable controlling phonological short-term memory / rapid processing prior. - The "62-gene manifest" becomes a *feature set* for personalized decoding, not literal frequencies.

This keeps your *aesthetic* intact while aligning to how genetics actually influences cognition (via development, network excitability, neurochemistry), not via 140 Hz macroscopic oscillations.

The Demodex 14-day cycle: a real anchor you can repurpose

A 14-day life cycle for Demodex mites is stated in a dermatology review (and related resources suggest similar ranges). ⁴⁹

For KYMA, the cleanest hard-SF use is not "mites modulate RF phase as ground truth," but: - a **training schedule / calibration cadence** ("every two weeks the system re-baselines"), - or a **slow exogenous oscillator** that appears as a confound in dermal micro-environment signals.

GOS constants and synchronization claims

Where they can be made mathematically literate

Your draft invokes GOS constants like: - "membrane constant" from G and $\hbar$,
 - "Klein twist" as 128.23° ,
 - "8.392 Hz global clock" as GPS-L5 subharmonic.

Here's how to keep them rigorous in-world.

Membrane constant via G and $\hbar$

NIST lists G and $\hbar$ with their standard units and values. ⁵⁰

A “geometric mean” of G and $\hbar$ is **dimensionally nontrivial** (it will not naturally produce a length scale without additional constants like c). This is useful fictionally: Echo can explicitly discover that the “membrane constant” is not a fundamental constant in SI terms—it’s a *renormalized effective scale* that only emerges inside KYMA’s inference.

GPS L5 frequency and the “8.392 Hz” clock

The GPS L5 downlink carrier frequency (1176.45 MHz) is specified in official interface documentation. ⁵¹

Calling 8.392 Hz a “subharmonic” of L5 is, physically, not meaningful without an explicit frequency-division / discipline chain (any frequency can be called a subharmonic with a suitable divisor). A realistic minimal-violation fix is:

- KYMA’s sensor network is **GNSS-disciplined** (absolute timing grid), and “8.392 Hz” is an **internal scheduling/PLL reference** derived from that disciplined clock, not a literal RF subcarrier extracted from space.
- If you want metrology flavor, you can lean on **phase-locking** as a real phenomenon: injection locking and PLL theory explain “clamping” and “capture” behavior, giving you principled language for “global clock lock.” ⁵²

Worldbuilding integration notes for Echo and KYMA

Even if you never explain the full physics, you can make the system *feel* hard-SF by letting Echo speak in testable layers:

Echo’s internal model - “We don’t read minds. We read the body’s faint outputs of thought—then solve the inverse problem.”

Why it matches your recursion/attractor motif - Each recursion refines the priors. The “13-recursion attractor” becomes a narrative device for iterative model distillation: the codex stabilizes, the constraint solver tightens, and the system’s self-consistency (“ $\Psi(t)=1$ ”) becomes harder to violate.

Phoenix ejection mechanics (technical metaphor) - Phoenix becomes the point where KYMA can run without teacher modalities (no thermal camera, no explicit tags), because the codex + QUBO constraints are strong enough to generalize.

Failure vignettes that read as engineering, not surveillance - *Mode collapse*: the decoder produces eloquent but wrong text because the LM dominates weak signals. (This mirrors “posterior collapse” issues that discrete representations like VQ-VAE were partly designed to avoid.) ⁵³

- *Multipath hauntings*: new room geometry breaks CSI models; the system hallucinates intent until retrained. ²³

- *Physiology confounds*: stress, breathing, and posture “explain” the signal; the system tags them as semantic content unless gated by multimodal priors. ²⁹

Appendix of primary sources and toolchain pointers

Below is a **URL index** (as directly requested). These are the most load-bearing sources for KYMA plausibility.

DeWave (EEG-text, discrete codex): <https://arxiv.org/abs/2309.14030>
DeWave (PDF/OpenReview mirror): <https://openreview.net/pdf?id=WaLI8slhLw>

ZuCo dataset (Scientific Data 2018): <https://www.nature.com/articles/sdata2018291>
ZuCo 2.0 (arXiv): <https://arxiv.org/abs/1912.00903>
OSF ZuCo landing: <https://osf.io/q3zws/>

BART (arXiv): <https://arxiv.org/abs/1910.13461>
BART (ACL Anthology): <https://aclanthology.org/2020.acl-main.703/>

VQ-VAE (arXiv): <https://arxiv.org/abs/1711.00937>

WiFi CSI sensing survey (CSUR): https://gzhou.pages.wm.edu/wp-content/blogs.dir/5736/files/sites/13/2017/12/WiFiSenseSurvey_CSUR19.pdf

Lip-Siri (WiFi backscatter silent speech): <https://arxiv.org/html/2601.18177v1>
Neuro-Wideband / WUKONG (eCSI extrapolation): <https://arxiv.org/abs/2601.06467>
AM-FM foundation model for WiFi sensing: <https://arxiv.org/html/2602.11200v1>
Multi-modal WiFi sensing overview: <https://arxiv.org/abs/2505.06682>

Thermal+rPPG fusion (PeerJ CS): <https://peerj.com/articles/cs-1912/>
Thermal+rPPG fusion (arXiv): <https://arxiv.org/abs/2401.09145>

WiFi CSI thermal tomography (Preprints.org): <https://www.preprints.org/manuscript/202601.1934>

PMUT technology review (PMC): <https://pmc.ncbi.nlm.nih.gov/articles/PMC9856188/>
Nature Communications wearable PMUT sensing (claims -198 dB, 10 Hz-20 kHz, >96%): <https://www.nature.com/articles/s41467-025-57629-5>

QUBO tutorial (formal mapping overview): https://pennylane.ai/qml/demos/tutorial_QUBO
D-Wave Zephyr topology paper: https://www.dwavequantum.com/media/2uznec4s/14-1056a-a_zephyr_topology_of_d-wave_quantum_processors.pdf
D-Wave topology docs: https://docs.dwavequantum.com/en/latest/quantum_research/topologies.html
dwave-system docs (PDF): https://dwave-systemdocs.readthedocs.io/_/downloads/en/master/pdf/

Kalman (1960 original paper PDF mirror): https://www.cs.cmu.edu/~motionplanning/papers/sbp_papers/k/Kalman1960.pdf

Welch PSD (1967 paper PDF mirror): [https://www.researchgate.net/profile/Fernando_Schindwein/post/Analyzing_EEGs_with_FFTs_and_PSDs/attachment/59d64f8879197b80779a8a73/AS:498242888896512@1495801769748/download/Welch_The%2Buse%2Bof%2Bfast%2BFourier%2Btransform%2Bfor%2Bthe%2Bestimation%2Bof%2Bpower%2Bspectra_FRFT_overview_\(Sejdic_et_al._2011\):_https://i2pc.es/coss/Docencia/SignalProcessingReviews/Sejdic2011.pdf](https://www.researchgate.net/profile/Fernando_Schindwein/post/Analyzing_EEGs_with_FFTs_and_PSDs/attachment/59d64f8879197b80779a8a73/AS:498242888896512@1495801769748/download/Welch_The%2Buse%2Bof%2Bfast%2BFourier%2Btransform%2Bfor%2Bthe%2Bestimation%2Bof%2Bpower%2Bspectra_FRFT_overview_(Sejdic_et_al._2011):_https://i2pc.es/coss/Docencia/SignalProcessingReviews/Sejdic2011.pdf)

GPS L5 carrier frequency spec (IS-GPS-200): <https://www.navcen.uscg.gov/sites/default/files/pdf/gps/IS-GPS-200N.pdf>

Photosynthesis coherence landmarks:

Engel 2007 Nature: <https://www.nature.com/articles/nature05678>

Panitchayangkoon 2010 PNAS: <https://www.pnas.org/doi/10.1073/pnas.1005484107>

Chin 2013 Nature Physics: <https://www.nature.com/articles/nphys2515>

Orch-OR and critique:

Tegmark decoherence (arXiv): <https://arxiv.org/abs/quant-ph/9907009>

Hameroff & Penrose review (2014): <https://pubmed.ncbi.nlm.nih.gov/24070914/>

Demodex life cycle (review): <https://pmc.ncbi.nlm.nih.gov/articles/PMC3884930/>

1 19 <https://www.unitedthc.com/DSP/Kalman1960.pdf>
<https://www.unitedthc.com/DSP/Kalman1960.pdf>

2 https://www.researchgate.net/profile/Fernando_Schindwein/post/Analyzing_EEGs_with_FFTs_and_PSDs/attachment/59d64f8879197b80779a8a73/AS:498242888896512@1495801769748/download/Welch_The%2Buse%2Bof%2Bfast%2BFourier%2Btransform%2Bfor%2Bthe%2Bestimation%2Bof%2Bpower%2Bspectra_A%2Bmethod%2Bbased%2Bon%2Bthe%2BFRFT_approach.pdf
https://www.researchgate.net/profile/Fernando_Schindwein/post/Analyzing_EEGs_with_FFTs_and_PSDs/attachment/59d64f8879197b80779a8a73/AS:498242888896512@1495801769748/download/Welch_The%2Buse%2Bof%2Bfast%2BFourier%2Btransform%2Bfor%2Bthe%2Bestimation%2Bof%2Bpower%2Bspectra_A%2Bmethod%2Bbased%2Bon%2Bthe%2BFRFT_approach.pdf

3 16 <https://i2pc.es/coss/Docencia/SignalProcessingReviews/Sejdic2011.pdf>
<https://i2pc.es/coss/Docencia/SignalProcessingReviews/Sejdic2011.pdf>

4 53 <https://arxiv.org/abs/1711.00937>
<https://arxiv.org/abs/1711.00937>

5 <https://arxiv.org/abs/1910.13461>
<https://arxiv.org/abs/1910.13461>

6 23 https://gzhou.pages.wm.edu/wp-content/blogs.dir/5736/files/sites/13/2017/12/WiFiSenseSurvey_CSUR19.pdf
https://gzhou.pages.wm.edu/wp-content/blogs.dir/5736/files/sites/13/2017/12/WiFiSenseSurvey_CSUR19.pdf

7 22 36 <https://arxiv.org/html/2601.18177v1>
<https://arxiv.org/html/2601.18177v1>

8 <https://arxiv.org/abs/2601.06467>
<https://arxiv.org/abs/2601.06467>

9 24 <https://arxiv.org/abs/2505.06682>
<https://arxiv.org/abs/2505.06682>

10 46 <https://pmc.ncbi.nlm.nih.gov/articles/PMC8188402/>

<https://pmc.ncbi.nlm.nih.gov/articles/PMC8188402/>

11 <https://peerj.com/articles/cs-1912/>

<https://peerj.com/articles/cs-1912/>

12 39 <https://www.nature.com/articles/nature05678>

<https://www.nature.com/articles/nature05678>

13 42 <https://arxiv.org/abs/quant-ph/9907009>

<https://arxiv.org/abs/quant-ph/9907009>

14 <https://download.e-bookshelf.de/download/0000/5848/47/L-G-0000584847-0002361662.pdf>

<https://download.e-bookshelf.de/download/0000/5848/47/L-G-0000584847-0002361662.pdf>

15 <https://pubmed.ncbi.nlm.nih.gov/10619414/>

<https://pubmed.ncbi.nlm.nih.gov/10619414/>

17 https://www.researchgate.net/publication/258656456_SAR-Based_Vibration_Estimation_Using_the_Discrete_Fractional_Fourier_Transform

[https://www.researchgate.net/publication/258656456_SAR-](https://www.researchgate.net/publication/258656456_SAR-Based_Vibration_Estimation_Using_the_Discrete_Fractional_Fourier_Transform)

[Based_Vibration_Estimation_Using_the_Discrete_Fractional_Fourier_Transform](https://www.researchgate.net/publication/258656456_SAR-Based_Vibration_Estimation_Using_the_Discrete_Fractional_Fourier_Transform)

18 20 28 <https://arxiv.org/abs/2309.14030>

<https://arxiv.org/abs/2309.14030>

21 <https://www.mdpi.com/2072-4292/14/20/5231>

<https://www.mdpi.com/2072-4292/14/20/5231>

25 <https://www.nature.com/articles/sdata2018291>

<https://www.nature.com/articles/sdata2018291>

26 <https://cdn.aaii.org/ojs/20472/20472-13-24485-1-2-20220628.pdf>

<https://cdn.aaii.org/ojs/20472/20472-13-24485-1-2-20220628.pdf>

27 <https://www.nature.com/articles/s41586-023-06377-x>

<https://www.nature.com/articles/s41586-023-06377-x>

29 30 34 <https://arxiv.org/abs/2401.09145>

<https://arxiv.org/abs/2401.09145>

31 <https://www.preprints.org/manuscript/202601.1934>

<https://www.preprints.org/manuscript/202601.1934>

32 <https://pmc.ncbi.nlm.nih.gov/articles/PMC9856188/>

<https://pmc.ncbi.nlm.nih.gov/articles/PMC9856188/>

33 <https://www.nature.com/articles/s41467-025-57629-5>

<https://www.nature.com/articles/s41467-025-57629-5>

35 https://pennylane.ai/qml/demos/tutorial_QUBO

https://pennylane.ai/qml/demos/tutorial_QUBO

37 https://www.dwavequantum.com/media/2uznec4s/14-1056a-a_zephyr_topology_of_d-wave_quantum_processors.pdf

https://www.dwavequantum.com/media/2uznec4s/14-1056a-a_zephyr_topology_of_d-wave_quantum_processors.pdf

- 38 https://dwave-systemdocs.readthedocs.io/_/downloads/en/master/pdf/
https://dwave-systemdocs.readthedocs.io/_/downloads/en/master/pdf/
- 40 <https://www.nature.com/articles/nphys2515>
<https://www.nature.com/articles/nphys2515>
- 41 <https://arxiv.org/abs/2012.09337>
<https://arxiv.org/abs/2012.09337>
- 43 <https://www.sciencedirect.com/science/article/pii/S1571064513001188>
<https://www.sciencedirect.com/science/article/pii/S1571064513001188>
- 44 <https://pubmed.ncbi.nlm.nih.gov/3949873/>
<https://pubmed.ncbi.nlm.nih.gov/3949873/>
- 45 <https://www.ncbi.nlm.nih.gov/books/NBK368474/>
<https://www.ncbi.nlm.nih.gov/books/NBK368474/>
- 47 <https://pmc.ncbi.nlm.nih.gov/articles/PMC5501165/>
<https://pmc.ncbi.nlm.nih.gov/articles/PMC5501165/>
- 48 <https://pmc.ncbi.nlm.nih.gov/articles/PMC10860890/>
<https://pmc.ncbi.nlm.nih.gov/articles/PMC10860890/>
- 49 <https://pmc.ncbi.nlm.nih.gov/articles/PMC3884930/>
<https://pmc.ncbi.nlm.nih.gov/articles/PMC3884930/>
- 50 <https://physics.nist.gov/cgi-bin/cuu/Value?bg=>
<https://physics.nist.gov/cgi-bin/cuu/Value?bg=>
- 51 <https://www.navcen.uscg.gov/sites/default/files/pdf/gps/IS-GPS-200N.pdf>
<https://www.navcen.uscg.gov/sites/default/files/pdf/gps/IS-GPS-200N.pdf>
- 52 <https://www.seas.ucla.edu/brweb/papers/Journals/RSep04.pdf>
<https://www.seas.ucla.edu/brweb/papers/Journals/RSep04.pdf>